Stabilization of waves in exterior domains with boundary feedback

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Wave equation on unbounded domains

Consider $\mathcal{O} \subset \mathbb{R}^n$ a bounded open set with smooth boundary $\partial\mathcal{O}$, where n is an odd number. Denote $\Omega := \mathbb{R}^n \setminus \mathcal{O}$.

$$\begin{cases} \partial_t^2 u - \Delta u = 0, & \mathbb{R}_t^+ \times \Omega, \\ u = -\frac{\partial}{\partial \nu}(G u_t), & \mathbb{R}_t^+ \times \partial\Omega, \\ (u, \partial_t u)(0, x) = (f_1, f_2), & \Omega. \end{cases} \tag{Ψ}$$

Here ν is the exterior normal vector of $\partial\Omega$ and the feedback law is given by

$$G := (-\Delta)^{-1} : \dot{H}^{-1}(\Omega) \to \dot{H}_0^1(\Omega).$$

The energy space considered is $H := L^2(\Omega) \times \dot{H}^{-1}(\Omega)$.
Idea: Prove the exponential stabilization of the local energy

$$E_R(u)(t) = \frac{1}{2}\left[\|u(t)\|^2_{L^2(\Omega_R)} + \|\partial_t u(t)\|^2_{\dot{H}^{-1}(\Omega_R)}\right],$$

where $\overline{\mathcal{O}} \subset B_R$ and $\Omega_R := \Omega \cap B_R$.

Definition (Exterior geometric control condition)

Let $R, T_R > 0$ and $\omega \subset \partial\Omega$. We say that (ω, T_R) satisfies the exterior geometric control condition in B_R if every generalized bicharacteristic ray γ starting from $T_b^*(\mathbb{R}_+ \times \Omega_R)$ verifies one of the next conditions:

1. γ escapes from $\mathbb{R}_+ \times \Omega_R$ before time T_R, or
2. γ hits $\mathbb{R}_+ \times \omega$ between time 0 and T_R in a non-diffractive point.

Theorem (To be proved!)

Let $R, T_R > 0$ such that the pair $(\partial\Omega, T_R)$ verifies the Exterior Geometric Control Condition. Then $\exists C > 0, \alpha > 0$ such that for every $f \in H$ with support contained in B_R, the solution u of (Ψ) with initial data f satisfies

$$E_R(u)(t) \leq Ce^{-\alpha t} E(u)(0).$$

Geometric Control Condition

Definition (GCC)

Let $\omega \subset M$ be an non-empty open subset. Every generalized geodesic of M traveling at speed 1 meets ω in time $t \in (0, T)$.

Theorem

Assume (ω, T) satisfies GCC. Then, there exists $C > 0$ s.t for any $(v_0, v_1) \in H_0^1(M) \times L^2(M) =: \mathcal{E}$ and v solves Dirichlet wave equation on M, we have

$$\|(v_0, v_1)\|_{\mathcal{E}} \leq C \int_0^T \|\mathbf{1}_\omega \partial_t v(t)\|^2_{L^2(M)} dt.$$



C. Bardos, G. Lebeau, J. Rauch (1992)
Sharp sufficient conditions for the observation, control and stabilization of waves from the boundary

.

What do we know on unbounded domains?

$$\begin{cases} \partial_t^2 u - \Delta u + a(x)\partial_t u = 0, & \mathbb{R} \times \Omega, \\ u = 0, & \mathbb{R}_+ \times \Omega, \\ u(0) = f_1, \ \partial_t u(0) = f_2. \end{cases} \tag{E}$$

Exponential stabilization by means of the same microlocal assumption, Lax-Phillips semigroup and microlocal defect measures

Energy space: $H^1(\Omega) \times L^2(\Omega)$, $\omega = \text{supp}(a) \subset \Omega$



L. Aloui, M. Khenissi (2002)
Stabilisation pour l'équation des ondes dans un domaine extérieur

.

And with some boundary feedback?

$$\begin{cases} \partial_t^2 u - \Delta u = 0, & \mathbb{R}_+ \times \Omega, \\ \partial_n u + a(x)\partial_t u = 0, & \mathbb{R}_+ \times \partial\Omega, \\ u(0) = f_1,\ \partial_t u(0) = f_2, & \Omega, \end{cases} \tag{N_a}$$

Same as before, but with more geometrical work and the celebrated Bardos-Lebeau-Rauch Lifting Lemma since we have dynamics on the boundary

Energy space: $H^1(\Omega) \times L^2(\Omega)$, $\omega = \text{supp}(a) \subset \partial\Omega$.



L. Aloui (2002)
Stabilisation Neumann pour l'équation des ondes dans un domaine extérieur

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Pseudodifferential feedback: Bounded domain

$\overline{\Gamma_1} \cap \overline{\Gamma_0} = \emptyset$, $\text{int}(\Gamma_0) \neq \emptyset$, $\Gamma_1 \cup \Gamma_0 = \partial\Omega$. $G := (-\Delta)^{-1} : H^{-1}(\Omega) \to H_0^1(\Omega)$.

$$\begin{cases} \partial_t^2 u - \Delta u = 0, & \mathbb{R}_t^+ \times \Omega, \\ u = -\frac{\partial}{\partial \nu}(G u_t), & \mathbb{R}_t^+ \times \Gamma_0, \\ u = 0, & \mathbb{R}_t^+ \times \Gamma_1, \\ (u, \partial_t u)(0, x) = (f_1, f_2), & \Omega. \end{cases} \tag{1}$$

Exponential stabilization by GCC + Compactness-uniqueness.

Energy space: $L^2(\Omega) \times H^{-1}(\Omega)$.



C. Bardos, L. Halpern, G. Lebeau, J. Rauch, E. Zuazua (1991)
Stabilisation de l'équation des ondes au moyen d'un feedback portant sur la condition aux limites de Dirichlet

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Well-posedness

We denote $-\Delta$ by A, which is a strictly positive self-adjoint operator. In $\dot{H}^{-1}(\Omega)$ we have the following inner product

$$(v_1, v_2)_{\dot{H}^{-1}(\Omega)} := (A^{-1/2} v_1, A^{-1/2} v_2)_{L^2(\Omega)}.$$

Note that by the hhyperbolic energy estimate we have

$$\begin{aligned} \frac{d}{dt} E_R(u)(t) &= \int_{\partial\Omega} \partial_n G(\partial_t u) \left[(-\Delta)^{-1}\right] (\partial_t u) d\sigma \\ &= -\int_{\partial\Omega} \left|\partial_n \left[(-\Delta)^{-1}\right]\right|^2 d\sigma \leq 0, \end{aligned}$$

which justifies the chosen feedback. Hereafter we will use $U_a(t)$ to denote the semigroup associated to

$$\partial \begin{pmatrix} u \\ v \end{pmatrix} = A \begin{pmatrix} u \\ v \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & I \\ \Delta & 0 \end{pmatrix}.$$

Well-posedness

Consider the unbounded operator A_a which operates as before, with domain

$$D(A_a) = \{(f_1, f_2) \in H : \ (f_2, \Delta f_1) \in H, \ f_1 + \partial_\nu(Gf_2) = 0 \text{ on } \partial\Omega.\}$$

Proposition

A_a is a maximal dissipative operator.

Proof: By the energy estimate, A_a is dissipative. It suffices to prove that $\mathrm{Im}(A_a - I)$ is closed and dense in H which is classical using completeness of the space + Lax-Milgram Theorem, respectively. □

Hence by Hille-Yosida Theorem, A_a generates a contraction semigroup $(U_a(t))_{t\geq 0}$ on H. For $G \equiv 0$, A_a is an anti-adjoint operator and it generates an unitary group $(U(t)_{t\in\mathbb{R}})$.

Lax-Phillips semigroup

We define the spaces of outgoing and incoming data as

$$D_{+,R} = \{f \in H : U(t)f = 0 \text{ over } |x| \leq t + R,\ t \geq 0\},$$
$$D_{-,R} = \{f \in H : U(t)f = 0 \text{ over } |x| \leq -t + R,\ t \geq 0\}.$$

Consider H_0 as the completion of $(C_0^\infty(\mathbb{R}^n))^2$ with the $H^1(\mathbb{R}^n)$ norm and identify H as a subspace. We denote by $D_\pm$ to the previous spaces when $R = 0$. Define the extension operator

$$E : H \to H_0, \quad E = \chi R,$$

where R is the exntesion by reflection and $\chi \in C^\infty(\mathbb{R}^n)$ such that

$$\chi = \begin{cases} 0, & \text{over a compact subset contained in } \mathcal{O}, \\ 1, & \text{on a neighbourhood of } \overline{\Omega}. \end{cases}$$

Then $ED_+ = D_+$ and $ED_- = D_-$.

Lax-Phillips semigroup

For $t \geq 0$, we write

$$Z_a(t) = P^+ U_a(t) P^-,$$

where P^+ (resp. P^-) denotes the projection onto $D_+^\perp$ (resp. $D_-^\perp$). Then $Z_a(t)$ verifies

Proposition

We have

1. $Z_a(t)D_+ = Z_a(t)D_- = 0$.
2. $Z_a(t)$ *operates over* $K = D_+^\perp \cap D_-^\perp$.
3. $(Z_a(t))_{t\geq 0}$ *generates a semigroup over* K.

Proof. 1. By definition $Z_a(t)D_- = 0$. For the other, let $f \in D_+$ and $g \in D_-$. Then

$$(f, g) = (Ef, Eg) = 0,$$

that is $f \in D_-^\perp$ and therefore $P^- f = f$. Since $f \in D_+$, $U_a(t)f = U(t)f \in D_+$, we get that $Z_a(t) = 0$.

Proof of the Proposition

2. Follows from

Lemma

Let B_1, B_2 be two maximal dissipative operators over a Hilbert space H. Suppose that B_1 is an extension of B_2 and $\overline{D(B_2)} = H$. Then $B_1 = B_2$.

$+$ definition of A_a^* and $D(A_a^*)$, which yields

$$0 = (Z_a(t)f, g) = (f, U_a^*(t)g), \quad \forall g \in D_-.$$

3. First we note $Z_a(t)f \in D_-^\perp$ for $f \in D_-^\perp$. Let $f \in K$, then

$$\begin{aligned} Z_a(t)Z_a(s)f &= P^+U_a(t)P^+U_a(s)f = P^+U_a(t)U_a(s)f \\ &= P^+U_a(t+s)f \\ &= Z_a(t+s)f \end{aligned}$$

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Why Microlocal Analysis?

1. More generality: Accepts different type of operators and regularity on coefficients.
2. **Elliptic world:** Microlocal Carleman Estimates due to Hörmander's "subellipticity" property
3. **Parabolic world:** Spectral inequalities for different operators.
4. **Hyperbolic world:** Propagation of regularity, smallness and compactness.

Microlocal analysis!

Hörmander's idea: Given a Riemannian manifold M, the natural space to do analysis is T^*M instead of M. By Fourier transform,

$$\text{Differential Operators} \quad \subset \quad \text{Pseudodifferential Operators}$$

is equivalent to

$$\text{Homogeneous Polynomials} \quad \subset \quad \text{Symbols}$$

There's a bijection between ΨDOs $\Psi^m(M)$ and symbols $S^m(T^*M)$.

Given $A \in \Psi^m(M)$, the principal symbol is $[\sigma_A^{\text{full}}] \in S^m(T^*M)/S^{m-1}(T^*M)$ and defines a surjective homomorphism

$$\sigma : \Psi^m(M)/\Psi^{m-1}(M) \to S^m(T^*M)/S^{m-1}(T^*M).$$

Manifolds without boundary: Bicharacteristics

Given $P \in \Psi^m(M)$ with principal symbol p singularities live in $p^{-1}(0)$ and are carried by the Hamiltonian field,

$$H_p(x,\xi) = \sum_{j=1}^{n} \frac{\partial p}{\partial \xi_j}\frac{\partial}{\partial x_j} - \frac{\partial p}{\partial x_j}\frac{\partial}{\partial \xi_j}.$$

The bicharacteristic curve of p is a maximal solution $\gamma : I \to T^*M$ such that

$$\dot{x}_j = \frac{\partial p}{\partial \xi_j}, \quad \dot{\xi}_j = -\frac{\partial p}{\partial x_j}, \quad (x(0),\xi(0)) = (x_0,\xi_0).$$

Projection $T^*M \ni (x,\xi) \mapsto x \in M$ are geodesics.

Manifolds without boundary: Wavefront set

Main idea: Study the propagation of some property as regularity (wavefront set) and compactness (microlocal defect measure).
Given $u \in \mathcal{D}'(M)$, we say that $(x_0, \xi_0) \notin \mathrm{WF}(u)$ iff there exists a neighbourhood of $V(x_0)$ and a conical neigh. $W(\xi_0)$ s.t $\varphi \in C_0^\infty(V)$ one has

$$\forall k \in \mathbb{N}, \exists C_k > 0 : |\widehat{\varphi u}(\xi)| \leq C_k(1 + |\xi|)^{-k}, \quad \forall \xi \in W.$$

This notion not only gives the singularity, it also shows the directions where it propagates.

Theorem (Hörmander 1971)

Let $P \in \Psi^m(\mathbb{R}^n)$, classical with real principal symbol. Let $u \in \mathcal{D}'(\mathbb{R}^n)$ s.t $Pu \in C^\infty(\mathbb{R}^n)$ and let γ be a bicharacteristic curve of P. Then

$$\gamma \subset \mathrm{WF}(u), \quad \text{or} \quad \gamma \cap \mathrm{WF}(u) = \emptyset.$$

Manifolds without boundary: Microlocal Defect Measures

Theorem (Gérard 1991)

Let $\{u_n\}_{n\in\mathbb{N}} \subset L^2_{loc}(M)$ weakly converging to 0. Then there exists a subsequence, a Radon measure $\mu \in \mathcal{M}(S^*M)$ such that for every $A \in \Psi^0(\Omega)$ with principal symbol $\sigma \in T^*M$, we have

$$(A(\chi u_n), \chi u_n)_{L^2(\Omega)} \to \int_{S^*M} \sigma(x,\xi) d\mu,$$

for every $\chi \in C_0^\infty(M)$ such that $\chi = 1$ in $\Pi_x(\mathrm{supp}(\sigma))$.

In particular, $\mu = 0$ in S^*M iff $u_n \to 0$

Theorem

Let $P \in \mathrm{Diff}^m(M)$ and $\{u_n\}_{n\in\mathbb{N}} \subset L^2_{loc}(M)$ weakly converging to 0 with MDM μ. TFAE,

1. $Pu_n \to 0$ *in* $H^{-m}_{loc}(\Omega)$,
2. $\mathrm{supp}(\mu) \subset \{(x,\xi) \in S^*M : \sigma_m(P)(x,\xi) = 0\}$.

An example: Bicharacteristics

For the wave operator $\Box = \partial_t^2 - \Delta$, the principal symbol is given by $p(\tau, \xi) = \tau^2 - |\xi|^2$. Bicharacteristics satisfies

$$\dot{t} = 2\tau, \quad \dot{x} = -2\xi, \quad \dot{\tau} = \dot{\xi} = 0,$$

so with the initial condition $(t_0, x_0, \tau_0, \xi_0) \in T^*\mathbb{R}^{1+n}$ such that $\tau_0^2 = |\xi_0|^2$, we get the bicharacteristic curve

$$\gamma(s) = (t_0 + 2s\tau_0, x_0 - 2s\xi_0, \tau_0, \xi_0), \quad s \in \mathbb{R}.$$

The projection $(t, x; \tau, \xi) \to (t, x)$ is called rays of the geometric optics.

An example: Propagation of regularity

Consider $u \in L^2((0,T)\times\Omega)$, $\omega \subset \Omega$ and

$$\begin{cases} \Box u = \partial_t^2 u - \Delta u = 0, & (0,T)\times\Omega, \\ \partial_t u \in L^2((0,T)\times\omega). \end{cases}$$

Then $u \in H^1_{\text{loc}}((0,T)\times\omega)$. Indeed,

$$u \in H^1_{\text{loc}} \iff \mathrm{WF}_1(u) = \emptyset \quad \wedge \quad \partial_t u \in L^2 \iff \mathrm{WF}_0(\partial_t u) = \emptyset.$$

Microlocal regularity: $\mathrm{WF}_{s-m}(Pu) \subset \mathrm{WF}_s(u) \subset \mathrm{Char}(P) \cup \mathrm{WF}_{s-m}(Pu)$.
Therefore intersecting,

$$\mathrm{WF}_1(u) \subset \mathrm{Char}(\Box) \cap \mathrm{Char}(\partial_t) = \emptyset,$$

we get $\mathrm{WF}_1(u) = \emptyset$, which gives $u \in H^1_{loc}((0,T)\times\omega)$.

An example: Why GCC is so important

Suppose that GCC holds and

$$\begin{cases} \Box u = 0, \quad (0,T)\times\Omega, \\ u = 0, \quad (0,T)\times\partial\Omega, \\ u \in L^2((0,T)\times\Omega), \\ \partial_t \in L^2((0,T)\times\omega). \end{cases}$$

By previous example $u \in H^1_{\text{loc}}((0,T)\times\omega)$. Take $\rho_0 = (t_0, x_0; \tau_0, \xi_0) \in (0,\varepsilon)\times\Omega\times\mathbb{R}^{1+n}\setminus\{0\}$, by GCC, the bicharacteristic γ_0 issued from ρ_0 enters $(0,T)\times\omega$, where $u \in H^1((0,T)\times\omega)$. By Hörmander's theorem, $\gamma \subset \mathrm{WF}_1(u)$, so $u \in H^1((0,T)\times\Omega)$

Manifolds with boundary

Denote by $M = \mathbb{R} \times \Omega$ and write $p(t, x, \tau, \xi) = \tau^2 - |\xi|^2$ for the principal symbol of $\Box = \partial_t^2 - \Delta$,

$$\text{Char}(\Box) = \Sigma = \{(t, x, \tau, \xi) \in T^*\overline{M} : p(t, x, \tau, \xi) = 0\},$$

the characteristic variety and $T_b^*M = T^*M \cup T^*\partial M$ the cotangent space at boundary. Consider the following natural projection

$$\pi : T^*\overline{M} \to T_b^*M, \quad \pi(t, x, \tau, \xi) = \begin{cases} (t, x, \tau, \xi), & (t, x) \in M \\ (t, x, (\tau, \xi)\,|_{T_{(t,x)}\partial M}), & (t, x) \in \partial M. \end{cases}$$

Manifolds with boundary: Generalized bicharacteristics

Definition

We define the subsets $\mathcal{E}, \mathcal{G}$ and $\mathcal{H}$ of $T^*\partial M$ as

$$\mathcal{E} = \{q \in T^*\partial M : \#\{\pi^{-1}(q) \cap \Sigma\} = 0\},$$
$$\mathcal{G} = \{q \in T^*\partial M : \#\{\pi^{-1}(q) \cap \Sigma\} = 1\},$$
$$\mathcal{H} = \{q \in T^*\partial M : \#\{\pi^{-1}(q) \cap \Sigma\} = 2\}$$

$\mathcal{E}, \mathcal{G}$ and $\mathcal{G}$ are, respectively, the set of elliptic, glancing and hyperbolic points.

- Let $\rho \in \mathcal{G}$, ρ is said strictly diffractive if and only if $H_p^2(y_n)(\rho) > 0$. The set of strictly diffractive points is denoted by $\mathcal{G}_d$.
- Let $q \in T^*\partial M$, q is said a non diffractive point if and only if $q \in \mathcal{E} \cup \mathcal{H}$ or $q \in \mathcal{G}$ and $\forall \varepsilon > 0$, there exists $s \in (-\varepsilon, \varepsilon) \setminus \{0\}$ such that $\gamma(s) \notin T^*\mathbb{R}^{n+1}\,|_{\overline{M}}$ where γ is the bicharacteristic curve of $\Box$ starting at a point $\tilde{q} = \pi^{-1}(q) \cap \Sigma$.

Boundary interaction

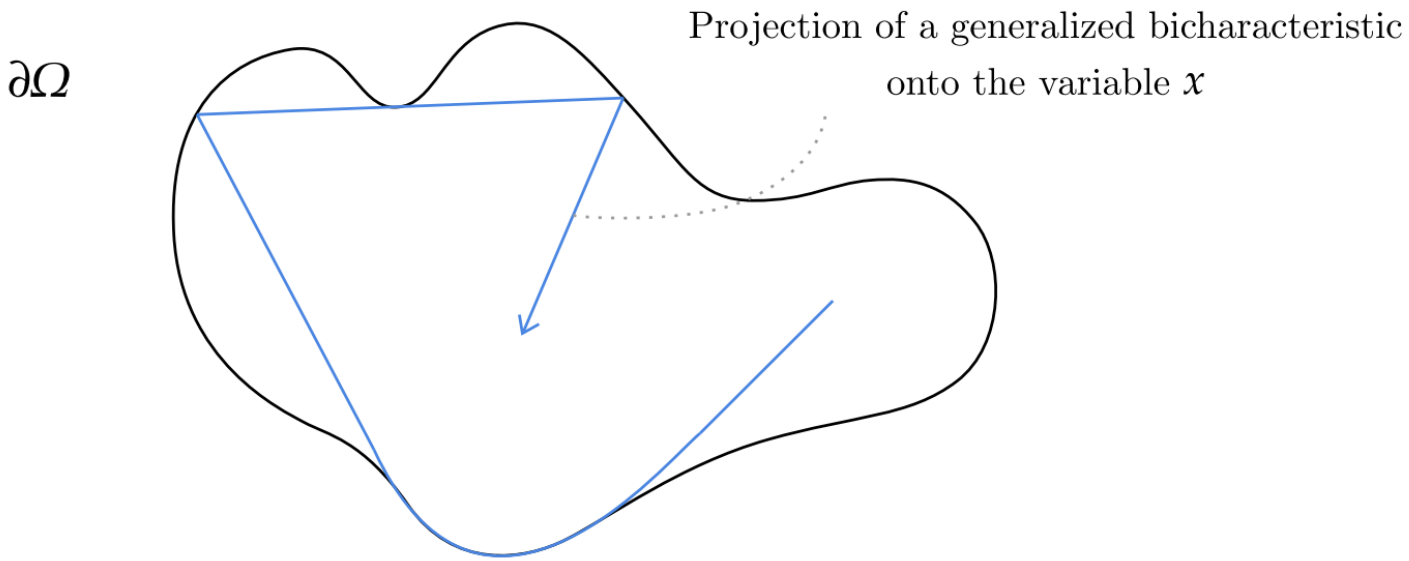


Figure 1.1: Generalized bicharacteristics

C. Loyola (2025)
Stabilisation, contrôle et prolongement unique pour quelques équations aux dérivées partielles nonlinéaires dispersives (Thèse de Doctorat en Sciences)

Manifolds with boundary: propagation results

Theorem (Melrose-Sjöstrand)

Let u be a distribution over M, extendable such that $\Box u \in C^\infty(\overline{M})$ and over ∂M we have $u = 0$ or $\partial_n u = 0$. If γ is a generalized bicharacteristic ray of $\Box$, then we have:

1. $\mathrm{WF}_b(u) \subset \Sigma_b = \pi(\Sigma)$,
2. $\gamma \subset \mathrm{WF}_b(u)$ *or* $\gamma \cap \mathrm{WF}_b(u) = \emptyset$.

Theorem (Lebeau)

Denote by $G(s)$ the generalized bicharacteristic flow and by μ the MDM. Suppose that ∂M doesn't have a contact point of infinite order. Then there exists $C > 0$ such that for all $s \geq 0$, $\mu^+(G(s)B) \leq C\mu^+(B)$ for every Borelian B on $\pi(\Sigma)/\mathbb{R}^*_+$.

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Lifting lemma

Theorem (Lifting lemma, Neumann case)

Let $q = (t_0, x_0'; \tau_0, \xi_0') \in T^*\partial M$ be a non-diffractive point and $I \times \omega$ a neigh. of $(t_0, x_0 = (x_0', 0))$ in M. Let $\{u_k\}_{k\in\mathbb{N}}$ be a solution of (N_a) verifying $u_k \rightharpoonup 0$ in H^1_{loc} and μ the associated MDM. Suppose that

$$u_k \mid_{\partial M} \to 0 \text{ in } H^1(I \times \partial\omega) \text{ and } \partial_n u_k \mid_{\partial M} \to 0 \text{ on } L^2(I \times \partial\omega).$$

Then $\mu \equiv 0$ around q.

In our case: Main problem is to replicate ideas with less regularity assumptions.

Stabilization result: Aloui's Neumann case

It's not hard to prove that $U_a(t)f = Z_a(t)f$ over B_R. It suffices to prove $\|Z_a(T)\| < 1$ for some $T > 0$.

Lemma

For $T = T_R + 9R$, $\|Z_a(T)\| < 1$.

"Proof": By contradiction, $\exists\{f_k\}_{k\in\mathbb{N}} \subset K : \|f_k\| = 1$ and $\|Z_a(T)f_k\| \to 1$. Since $\|U_a(t)f_k\| \leq \|f_k\|$,

$$\Pi_x(U_a(t)f_k) := u_k \text{ is bounded in } H^1_{\text{loc}} \text{ with } \mu \text{ as MDM}$$

then we need

Lemma

$\mu \equiv 0$ *on* $T^*(K(T'))$ *where* $T' = T_R + 3R$ *and*

$$K(T') = \{(t,x) \in \mathbb{R}_+ \times \Omega : |x| \leq t - T' + R,\ t \geq T'\}.$$

Lemma's proof: using EGCC

Let γ the bichar. with $q \in T^*(K(T'))$ as initial point.

Case 1: γ does not pass $\partial\Omega$ or it hits $\partial\Omega$ after time $t_0 > 2R$

Denote by μ_0 the MDM attached to $\Pi_x(U(t)f_k)$. We have $\mu = \mu_0$ on $\{\gamma(t) : 0 \leq t \leq t_0\}$. By Lax-Phillips semigroup, $\mu_0 = 0$ around $\gamma(t_1)$, where $t_1 < t_0$ is s.t $\gamma(t_1) \in T^*(K(2R))$. By propagation theorem, $\mu = 0$ around q.

Case 2: γ hits $\partial\Omega$ for $t_0 \leq 2R$.

Note that $\gamma(T'), \gamma(t_0) \in B_R$ and $T' - t_0 > T_R$. By EGCC, γ passes through $\omega \times [t_0, T]$ on a non-diffractive point $\tilde{q}$. By energy estimates,

$$(u_k \mid_{\partial M}, \partial_n u_k \mid_{\partial M}) \to (0,0) \; H^1(\omega \times [t_0, T]) \times L^2(\omega \times [t_0, T]).$$

By Lifting Lemma, $\mu = 0$ around $\tilde{q}$, and by propagation theorem $\mu = 0$ around q.



Back to previous lemma

The first part comes from the following domain of dependence result

Lemma

$\forall g \in D_{-}^{\perp}$, $\|Z_a(2R)g\| \leq 2\|g\|_{5R}$.

and it implies $\|Z_a(T)f\| = \|f\| = 1$. Then by

Lemma

$G_T = \{f \in K : \|Z_a(T)f\| = \|f\|\} = \{0\}$

we achieve a contradiction.

Sketch of the proof: We need to prove 1. $\dim(G_T) < +\infty$, 2. A operates over G_T and 3. A doesn't have eigenvalues. Hence $f \in K$ satisfies an elliptic equation with analytics coefficients. Hence f is analytic, and by Dirichlet boundary condition, $f \equiv 0$.

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Some extensions

1. EGCC forces the damping mechanism. It is possible to have mixed boundary conditions? (i.e, $u = 0$ on $\Gamma_1 \subset \partial\Omega$)
2. There's an extension in the Dirichlet case with a decreasing non-linearity with polynomial growth f and $a \in L^\infty_+(\Omega)$ with compact support,

$$\begin{cases} \partial_t^2 u - \Delta u + a(x) f(\partial_t t u) = 0, & \mathbb{R}^+_t \times \Omega, \\ u = 0, & \mathbb{R}^+_t \times \partial\Omega, \\ (u, \partial_t u)(0, x) = (\varphi_1, \varphi_2), & \Omega. \end{cases}$$

Is it possible to extend to the Neumann and Pseudodifferential case?



M Daoulatli (2007)
Local energy decay for the nonlinear dissipative wave equation in an exterior domain

Thanks for your attention!